STAFF REVIEW · H446 313 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Copies ACC contents to the specified mailbox.

2. Address of the memory location being accessed.

3. Maximum CPU slice before RR rescheduling.

4. 10_2: sum0, carry1.

I Do — source worked steps

1. Predict first — PC holds address of next instruction, not the instruction contents.

2. Explain and check — Explain the reason for the result.

We Do — guided CHECK

1. Predict first
Arrival-order service makes B wait three CPU units.
Explain and check
Explain the reason for the result.

You Do — source mark schemes

1. ACC4 [1], A4 [1], ACC2 [1], B2 [1], LDA A→4 [1], SUB B→2 and OUT2 [1].

2. FCFS A,B,C waits 0,4,6 mean10/3 [3]; SJF C,B,A waits 0,1,3 mean4/3 [3].

3. Clock frequency matters [1], cache misses stall [1], cache may reduce latency [1], pipeline overlaps stages [1], branches can flush/stall [1], performance depends on bottlenecks [1].

Exit Ticket — source answer

1. False. LDA X loads; STA X stores.